

Vedant Desai

(248) 704 – 4852 | vedantde@umich.edu | github.com/Verdent06 | linkedin.com/in/vedantde06

EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0

- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Discrete Math, Calc III

EXPERIENCE

Lyndbrook Capital

Data Engineering Consultant

Feb 2026 – Apr 2026

Remote

- Built a Review Velocity scoring algorithm that proxied fleet expansion and operational scale from public compliance data, filtering 800 acquisition targets down to a 280-lead qualified shortlist — a 35% precision rate against the fund's revenue criteria.
- Contracted pre-LOI to build acquisition intelligence for a boutique search fund targeting water utility operators; aggregated EPA ECHO and MassGIS regulatory data into a unified PWSID entity database and delivered 800+ validated Day-1 acquisition targets within the engagement window.

CaseStudyPrep.AI

Software Engineer Co-op (Voice AI)

Dec 2025 – May 2026

Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

Michigan Data Consulting (MDC)

Data Engineer — Michigan Campaign Finance Network

Jan 2026 – May 2026

Ann Arbor, MI

- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Scoped technical delivery directly with MCFN stakeholders as the only engineer on contract — from ingestion pipelines through REST endpoints on EC2 — with no backend team to share infrastructure, API design, or deployment ownership.

PROJECTS

Granular Synthesizer Plugin | C++, JUCE, VST3, AU

github.com/Verdent06/granular-synth

Real-time audio DSP plugin built from scratch in C++/JUCE — sine oscillator through full granular engine with lock-free threading and release-ready VST3/AU binaries

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio parameter delivery through a hand-rolled lock-free SPSC FIFO so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.
- Configured CMake to compile VST3 and AU bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86_64) for native Apple Silicon and Intel — and audited every `processBlock()` path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before release builds.

TECHNICAL SKILLS

Languages Python, C++, TypeScript, SQL
Frameworks Flask, Angular, RxJS, JUCE
Cloud & Infrastructure AWS (EC2)